

RESEARCH INTERESTS

Human-Computer Interaction • Extended Reality (XR/VR/AR) • Tangible User Interfaces • AI-Enhanced Healthcare

EDUCATION

- University of Calgary

Bachelor of Science, Software Engineering

Relevant Coursework: Machine Learning, Computer Vision, Human-Computer Interaction, Algorithm Design, Data Structures, Software Architecture, Statistics, Linear Algebra

Calgary, AB

September 2021 – May 2026

EXPERIENCE

- Human-X Interaction Lab

Software Engineering Researcher

Published workshop paper at IEEE VR 2025 on tangible user interfaces for XR

Designed end-to-end research pipeline: Unity/C# XR apps, AWS EC2, REST APIs, web analytics

Leading research on novel interaction paradigms for mixed reality systems

Implemented hydrogen integration into natural gas pipelines using digital twin and AI-driven leak detection to improve safety, efficiency, and sustainability.

Calgary, AB

January 2025 - Present
- Cenovus Energy

Software Engineer Intern - Data Solutions

Built Ruby on Rails app replacing manual workflows, saving \$20M+ annually

Developed automation tools cutting processing time by 90%

Created low-latency data pipelines with Dagster, enhancing reliability and integration across teams.

Co-led internal dev community, running workshops on modern frameworks and AI to upskill early-career employees.

Calgary, AB

January 2024 - December 2024
- Arcurve

Software Development Intern

Built full-stack features using .NET and Angular; developed Azure monitoring tools

Participated in agile Scrum processes, code reviews, and pair programming sessions

Calgary, AB

May 2025 - August 2025
- Vertical City

Software Deployment Intern

Deployed Ubuntu 20 software for elevator/lobby screens across multiple properties

Documented installation processes; collaborated with Property Support Lead

Calgary, AB

July 2023 - August 2023
- JA Southern Alberta

Business Liaison (Part-time)

Mentored student companies; teams won 2 city and 1 national Company of the Year awards

Calgary, AB

October 2021 – March 2024

PEER-REVIEWED PUBLICATIONS

- IEEE Virtual Reality Conference 2025

TangibleMoments: Embedding XR Memories onto Physical Objects

Novel tangible user interface design for XR systems

Introduced new paradigms for physical-digital memory capture and re-visitation in virtual environments

April 2025

[Publication Link](#)
- STEM Fellowship Journal

Quantifying Childhood Trauma: Causal ML Approaches to Mental Health

Applied causal inference techniques to BRFSS dataset

Revealed nuanced relationships between ACEs and adult mental health trajectories

July 2024

[Publication Link](#)

HONORS AND AWARDS

- University of Calgary Innovation Grant (\$10,000, 2024) - BraVRy Mental Health Platform

Hack the Change Best Use of AI (2025) - AI-powered surveillance monitoring application for transit safety

Georgia Tech HackGT 2nd Place (2025) - Mixed reality museum to re-experience memories

IEEE SIGHT 1st Place (2023) - Accessibility-focused AR communication system

NeurAlbertaTech natHacks 2nd Place (2023) - Virtual Reality therapy platform

HackMedTech 3rd Place (2023) - Healthcare innovation project

IEEE VR Publication (2025) - Prestigious conference on immersive technology

STEM Fellowship Publication (2024) - Peer-reviewed journal publication

LEADERSHIP & COMMUNITY IMPACT

- **Hack the Bias** Calgary, AB
Founder & Lead Organizer *April 2025 – Present*
 - Founded Calgary's largest 36-hour beginner-friendly hackathon addressing AI bias
- **Robogals UCalgary** Calgary, AB
VP Special Projects (Current) — VP Finance & Operations (Previous) *August 2023 – Present*
 - Lead STEM education initiatives for underrepresented communities with 50+ volunteers
 - Helped run workshops for remote Indigenous communities to expand STEM access
 - Secured funding through grants; developed robotics curriculum with social impact focus
- **Schulich Ignite** Calgary, AB
Programming Mentor *September 2021 – May 2023*
 - Mentored high school students in Python and programming fundamentals through weekly sessions
- **Math Tutoring** Calgary, AB
Mathematics Tutor *January 2020 – May 2023*
 - Provided one-on-one and group math tutoring through YMCA and EquaLearn education programs

SELECTED PROJECTS

- **Aegis - Live AI Transit Security Monitoring** [GitHub](#)
FastAPI, AWS Bedrock, AWS EC2, Next.js *Hack the Change 2025 - Best Use of AI*
 - AI-powered real-time public transit surveillance
- **BraVRy - VR Exposure Therapy** [YouTube](#)
Meta Quest 2, Unity, AWS, OpenAI *natHacks 2023 - 2nd Place*
 - VR therapy for anxiety with biotracking; pitched at Inventures 2024, won UCalgary grant
- **Hall of Us - AI Mixed Reality Museum** [Devpost](#)
Quest 3, AI Scene Reconstruction, Computer Vision *HackGT 2025 - 2nd Place*
 - Transformed 2D memories into immersive 3D exhibits with AI-powered navigation
- **AutoLeARn - AR Car Maintenance** [GitHub](#)
Groq AI, Perplexity, Unity *TreeHacks 2025*
 - AR platform with speech/gesture controls for automotive repair guidance
- **PulseBot - Health Chatbot** [GitHub](#)
OpenAI, RAG, Chroma *HackMedTech 2025*
 - AI assistant for cardiovascular health using vectorized medical research and machine learning for ECG

TECHNICAL EXPERTISE

- **Research Tools:** PyTorch, TensorFlow, scikit-learn, R, Jupyter, LaTeX, Qualtrics
- **XR Development:** Unity, OpenXR, WebXR, MRTK, Meta SDK
- **Programming:** Python, C++, C#, JavaScript, Ruby, Java, SQL
- **Cloud & Systems:** AWS (EC2, Lambda, SageMaker), Docker, Kubernetes, CUDA, distributed computing
- **Data Science:** Pandas, NumPy, Snowflake, Databricks, statistical analysis, causal inference

REFERENCES

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